

Han Yan

Phone: +86 18019239175 | Email: k4rtol@163.com | WeChat: K4RTOL

Education

Australian National University (ANU) QS #34 — Master of Computing

Key courses: Computer Vision, Computer Graphics, Game Development, (Statistical) Machine Learning, Advanced Algorithms, SQL Databases.

Donghua University (211) — Bachelor of Functional Materials

Projects (<https://k4rto.com/>)

- **Smart OCR Hub — Customs Declaration Auto-Recognition & Structured Extraction (Python / Vision-LLM)**

Responsibilities & Contributions:

- **Dual-path OCR architecture:** primary path runs qwen3-vl-plus full-page OCR + LLM field merging; DocLayout-YOLO + PaddleOCR-VL-1.5 serves as local fallback so the service stays up when the API is unreachable.
- **Multi-GPU cluster scheduling:** vLLM instances + Redis state bus + SSE for live GPU utilization and alert push, supporting batch-concurrent multi-document processing.
- **End-to-end MLOps:** MLflow experiment tracking + human review loop + continuous fine-tuning, giving every model iteration an audit trail.
- **Stack:** FastAPI · SvelteKit · PostgreSQL · Redis · vLLM · MLflow.

- **AURA — Enterprise Multi-Agent Collaboration Platform (Python / SvelteFlow)**

Responsibilities & Contributions:

- **Blueprint orchestration engine:** visual DAG editor built on SvelteFlow turns SOPs into reusable agent workflows, with parallel dispatch, rate-limiting, version snapshots, and replay.
- **Multi-scenario adaptation:** a unified surface for meetings (live transcription + auto-dispatched minutes), projects (blueprint execution + RAG archival), and management (permission grouping + full audit trail).
- **Real-time collaboration core:** Socket.IO + Redis pub/sub + Yjs collaborative editing, supporting multi-user multi-Agent parallel tasks on a single screen.
- **Stack:** FastAPI · SQLAlchemy 2.0 · PostgreSQL · Alembic · SvelteKit 5 · MLflow · Open WebUI deep customization.

- **Human Fall Pose Detection & Audio Alert System (Python) — Joint project with a Western Australia company**

Responsibilities & Contributions:

- **Vision algorithm development:** customized YOLOv7 model for real-time human keypoint detection and pose analysis; structural optimization for accurate fall recognition.
- **Project Management (PM):** led project planning and module assignment, using Jira for team task tracking and progress visibility.

Highlight: Presented at the ANU Techlauncher showcase.

Details: [CAM FALL Project](#)

- **Game Engine Feature Development (C++)**

Responsibilities & Contributions:

- **Post-processing — Color Grading:** implemented flexible color grading via Look-Up Table (LUT) in the fragment shader, noticeably improving image quality.
- **Physics system improvements:** in-air control for character jumps and wall collisions, enhancing character-environment interaction realism.
- **Toolchain development:** leveraged the Piccolo engine's reflection system for dynamic property display/editing, enabling data-driven design and improving engine flexibility.

- **Vulkan-based C++ Grass Physics Rendering (C++)**

Responsibilities & Contributions:

- **Grass-blade physics:** shader-generated geometry with Bézier-curve shape, producing natural recovery under force.
- **Multiple wind effects:** helicopter wind, custom natural wind, and strong wind simulations, increasing scene realism.
- **Rendering optimization:** distance-based culling + frustum culling, reducing compute load.

- **GoodGame Social Media App (Java/Android)**

Responsibilities & Contributions:

- **Persistence & sync:** Firebase Firestore data access layer with safe, fast user-data storage.
- **Real-time sync:** Firestore listeners for multi-user real-time sync, ensuring instant chat / status updates.
- **Advanced search:** multi-field, multi-condition queries to improve search UX.
- **Custom UI components:** card view, expandable list, etc., improving visual quality and usability.

Internships & Work Experience

ByteDance | Junior Front-end Game Developer

2020.08 – 2020.10

- Built touch-based mini-games with Cocos Creator; owned UI/level design and authored game-design documents.

Shanghai Byron Culture Media Co. | Product Manager

2022.09 – 2022.12

- **Aliyun Apsara Conference:** led AR navigation and "Cloud Mascot" match-3 mini-game design and product documentation.
- **Shanghai Metro Guilin Road project:** owned the legal-culture wall content and AR interactive game design.

Shanghai Lingjing Hudong / Shanghai Lingdong Hudong | Product Manager

2021.03 – 2022.08

Owned WeChat mini-game data operations, driving UX improvements and retention.

Featured projects:

- **Rescue Town:** via WeChat Optimized Channel — 3M total users, retention from 15% → 27%, 500K three-month revenue.
- **Mini World Stellar Drift:** Mini World official-licensed IP — peak DAU 500K, ~100K monthly revenue across channels.
- **Survival Frontier · Mini Glacier · Young Knight:** retention and key-metric improvements.

Lingjing Oasis × Baidu Xirang × FengYuZhu — "Metaverse Spring Festival" event:

- Owned launch event planning for Lingjing Oasis × FengYuZhu, including the launch deck.
- Designed product prototypes, wrote product documentation, and managed R&D progress for the "Metaverse Spring Festival" project.
- Designed the warm-up campaign: 20K+ pre-registrations within 48 hours of launch.
- Three-day campaign reached 10K+ participants, total prize pool 1M — became Baidu Xirang's officially featured case and China's first large-scale metaverse event.